

Rachel Baker-Ramos

Computer Science PhD Student

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RESEARCH INTERESTS

I'm a researcher and designer in **educational technology** and **community-based design**, with a focus on the intersections of culturally-revitalizing education, AI for low-resource languages, and game-based learning. I bring robust experience in human-centered design, participatory research, and project leadership, with **two rigorous design degrees** and **7+ years co-designing with communities**. My work is informed by my multidisciplinary background and Latine heritage, aiming to provide equitable, engaging, and culturally affirming educational experiences to K-12 students.

EDUCATION

- 2026– **Georgia Institute of Technology**, *PhD Computer Science*
Advisors: Josiah Hester (Georgia Tech), Alex Cabral (M.I.T.)
★ **Stamps Doctoral Fellowship** ★ **Presidential Fellowship** (awarded, replaced by Stamps)
- 2022–24 **Georgia Institute of Technology**, *MS Human-Computer Interaction*
Thesis: *Kumu Connect: Co-designing Culturally-Relevant Generative EdTech with Hawaiian Immersion Educators*, Advisors: Josiah Hester, Judith Uchidiuno
★ **Best MS-HCI Thesis Award Finalist**
- 2014–19 **Carnegie Mellon University**, *Bachelor of Architecture (BArch)*, *add. major in Social & Political History*
★ **University Honors** ★ **College of Fine Arts Honors** ★ **Senior Leadership Award '2019**

WORK EXPERIENCE

- 2026– **Graduate Assistant**, *College of Computing, Georgia Institute of Technology*
Provide support for lab-wide projects (research protocol review, co-design and community-based design advising, IRB protocol support, paper writing & graphics). Write and prepare grant and award proposals. Supervise 3 MS/BS students in their independent studies and theses.
- 2024– **UX Design & Research Advisor**, *Ulu Lāhui Foundation*
Provide UX design & research support for educational & social impact technology projects. Prepare & write grant proposals, securing over 125k in funding from MIT Solve and aiEDU.
- 2024–26 **Research Scientist**, *College of Computing, Georgia Institute of Technology*
★ **College of Computing's 2025 Outstanding Research Scientist Award**
Led the Ka Moamoa Lab's Education & Public Interest HCI sub-group, managing 11+ MS/BS students across 7+ educational technology, public interest, and sustainability-minded initiatives. Authored 8 peer-reviewed publications on the intersections of culturally-relevant education, game-based learning, AI for low-resource languages, and community-based technology development. Wrote and prepared grant and award proposals, securing 475k funding from Google Research, Mozilla Foundation, and the NSF.
- 2023–24 **Graduate Research Assistant**, *College of Computing, Georgia Institute of Technology*
- 2023 **UX Design Fellow**, *U.S.C.I.S. Refugee, Asylum & International Operations Directorate*
- 2020–22 **Lead Graphic Designer**, *Veggie Mijas (Latina/x/e Environmental Justice Org)*
- 2019–22 **Project Designer**, *Caddis Collaborative (Architecture & Urban Design Studio)*

PUBLICATIONS

CONFERENCE PAPERS

- IDC 2026 **Kumu Connect: Culturally-Authentic AI Tools to Enable Educator Agency and Data Sovereignty for Hawaiian K-12 CS Education**
★ [Best Paper Honorable Mention](#)
Rachel Baker-Ramos, Manas Mhasakar, Viswak Raja, Evyn-Bree Helekahi-Kaiwi, William Gelder, Benjamin Carter, Rebecca Diego, Alex Cabral, Josiah Hester. 25th Annual ACM Interaction Design and Children Conference (IDC '26). (Acceptance rate: 23%)
- CHI 2026 **Noondawind: Co-Designed Dashboard for Indigenous Data Access and Environmental Policy Implementation**
★ [Best Paper Honorable Mention](#)
Julia Aileen McKenna, Gabriela Buraglia, Jahanvi Kolakaluri, **Rachel Baker-Ramos**, Samantha Carter, Joe Graveen, Jonathan Gilbert, James Rasmussen, Brandon Byrne, Darren Vogt, Josiah Hester, Kim Marion Suiseeya, Alex Cabral. ACM CHI Conference on Human Factors in Computing Systems (CHI '26)
- CHI 2026 **Whose Knowledge Counts? Co-Designing Community-Centered AI Auditing Tools with Educators in Hawai'i**
Dora Zhao, Hannah Cha, Michael J Ryan, Angelina Wang, **Rachel Baker-Ramos**, Evyn-Bree Helekahi-Kaiwi, Rebecca Diego, Josiah Hester, Diyi Yang. ACM CHI Conference on Human Factors in Computing Systems (CHI '26)
- RESPECT 2025 **Kumu Connect: Design Thinking for Place-Based Generative Educational Technology in Hawaiian Immersion Schools**
Rachel Baker-Ramos, Will Gelder, Leah Cho, Jahnvi Kolakaluri, Josiah Hester. 2025 ACM Conference on Research on Equitable and Sustained Participation in Engineering, Computing, and Technology (RESPECT '25).
- RESPECT 2024 **"Those don't work for us": An Assets-Based Approach to Incorporating Emerging Technologies in Viable Hawaiian Teacher Support Tools for Culturally Relevant CS Education**
William Gelder, **Rachel Baker-Ramos**, Ayoungh Cho, Jahnvi Kolakaluri, Judith Uchidiuno, Josiah Hester. 2024 ACM Conference on Research on Equitable and Sustained Participation in Engineering, Computing, and Technology (RESPECT '24).
- Koli Calling 2023 **Keeping Mindful of Modality: A Comparison of Computer Science Education Resources for Learning**
Michael J. Johnson, **Rachel Baker-Ramos**, Christopher L. Hovey, Betsy DiSalvo. 23rd ACM Koli Calling International Conference on Computing Education Research (Koli Calling '23)

POSTERS AND ABSTRACTS

- GoodIT 2026 **Still Warm: A Digital Game for Players to Address Climate Grief**
Rachel Baker-Ramos, Josiah Hester, Alex Cabral. ACM 6th International Conference on Information Technology for Social Good (GoodIT26). (Accepted)
- CHI 2025 **"I Would Never Trust Anything Western": Kumu (Educator) Perspectives on Use of LLMs for Culturally Revitalizing CS Education in Hawaiian Schools**
Manas Mhasakar*, **Rachel Baker-Ramos***, Ben Carter, Evyn-Bree Helekahi-Kaiwi, Josiah Hester. ACM CHI Conference on Human Factors in Computing Systems (CHI EA '25) *denotes equal contribution
- CHI PLAY 2024 **Enabling Restorative Hawaiian Futurism via a Tabletop Role-Playing Game and Digital Ha'i Mo'olelo (Storytelling)**
Rachel Baker-Ramos, Michael Parkin, Angelica Raza-Furtado, Alike S. Naihe, Jack Hobbs, Jeffrey Vierra, Aarom Ramirez, Chantel Rubin-Molina, Cheradean Kaaialii, Celeste Mason, Yaman Sangar, May Okihiro, Josiah Hester. 2024 ACM Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '24)

AWARDS & FELLOWSHIPS

- 2026 **Stamps Doctoral Fellowship**, *Georgia Institute of Technology*
Georgia Tech's most prestigious merit-based doctoral fellowship, created to attract exceptional scholars and support their development into leaders who will advance technology and improve the human condition.
- 2026 **Best Paper Honorable Mention**, *ACM Interaction Design and Children 25th Conference*
for "Kumu Connect: Culturally-Authentic AI Tools to Enable Educator Agency and Data Sovereignty for Hawaiian K-12 CS Education" (IDC 2026)
- 2025 **Outstanding Research Scientist Award**, *Georgia Institute of Technology, College of Computing*
- 2025 **Solver, Indigenous Communities Fellowship**, *MIT Solve*
Kumu Connect was selected from ~2,900 applicants (1% acceptance rate) for MIT Solve's Indigenous Communities Fellowship, a program for community-based solutions for and with Indigenous communities.
- 2023 **Civic Innovation Fellowship**, *Coding It Forward*
Selected from 2,500 student applicants (5% acceptance rate) for the Coding it Forward Summer Fellowship, a program for early-career technologists to explore civic & public interest technology.

SERVICE & VOLUNTEERING

- June 2026 **ACM Interaction Design and Children Conference**, *Student Volunteer*
- 2024 **UX of EdTech**, *Community Volunteer, Summer Cohort*
- 2023–24 **MS-HCI DEI Forum**, *DEI Forum Co-leader*
- 2020–23 **BoulderReads**, *Volunteer ESL Literacy Tutor*

PROJECTS & EFFORTS

- 2026– **Still Warm: A Game on Climate Grief**
- 2023– **Kumu Connect: A Culturally-Relevant CS Lesson Plan Generator**, *Role: Project Lead*
★ 23-24 Best MS-HCI Thesis Award Finalist ★ Google Academic Research Award
★ 2025 MIT Solve Indigenous Communities Fellowship solution
- 2024–25 **Co-Designing AI Bias Auditing with Educators in Hawai'i**, *Role: Community Engagement Lead*
- 2023–26 **Kaona: A Culturally-Relevant Educational Role-Playing Game and "Storyteller" Mobile App**,
Role: UX Designer & Game Artist
- 2023–24 **Internet Outage Detection and Analysis (IODA) Dashboard**, *Role: UX Designer*
- 2023 **Pickrick Project: An AR Installation on Atlanta Civil Rights History**, *Role: UX Designer*

GRANTWRITING & FUNDING

- 2026 **NSF Supplement Support Request: Translating Community-Driven Edge AI Data Systems into K–12 STEM Curricula for Rural Classrooms**, *National Science Foundation*
Project: SCC-IRG: Edge AI, Biofabrication, and Data Science for Disaster Preparedness and Community Wellbeing in Hawaii, Role: Personnel & Grant-writer
- 2025 **Indigenous Communities Fellowship**, *MIT Solve*
Project: Kumu Connect, Role: Senior Personnel & Grant-writer
- 2025 **Rural & Indigenous Community Catalyst Program**, *aiEDU*
Project: AI for Hawai'i Curricula Co-Creation Workshops & Resource Hub, Role: Senior Personnel & Grant-writer
- 2024 **Google Academic Research Award**, *Google Research*

Project: Kumu Connect: Community-informed AI Teacher Support Tools for Place-Based CS Curriculum in Hawaiian Schools, Role: Senior Personnel & Grant-writer

STUDENT MENTORING & REPORTS

2026–	Maha Sidi, MS-HCI , MS Thesis on "Atlanta Beltline & Public History"
2025–	Cheilota Uma, MS-HCI , Independent Study on "Kumu Connect: Code Generation" & MS Thesis on "Atlanta Beltline & Public History"
2024–	Ben Carter, BS-CS , Independent Study on "PLACES Education", Student Assistantship on "Kumu Connect" & BS Thesis on "Future Hawaiian AI"
2026	Dharmik, MS-CS , Independent Study on "Kumu Connect: RAG Experiments"
2026	Hanna Fu, MS-HCI , Independent Study on "Kumu Connect: Code Generation"
2026	Akanksha Tanwar, MS-CS , Independent Study on "Kumu Connect"
2025–26	James Kemerait, BS/MS-CS , Research Assistantship on "Kilo Tech"
2025–26	Isabella Sirianni, MS-HCI , MS Thesis on "Climate Investigators"
2025–26	Theresa Antony, MS-HCI , Independent Study on "Climate Games" & MS Thesis on "Climate Investigators"
2025	Viswak Raja, MS-HCI , Graduate Research Assistantship on "Kumu Connect"
2025	Christine Taylor, MS-HCI , Independent Study on "Noondawind: Bilingual Dashboards"
2024–25	Manas Mhasakar, MS-CS , Graduate Research Assistantship on "Kumu Connect"
2024–25	Jingxuan Wang, MS-HCI , Independent Study & Graduate Research Assistantship on "Elections Project"
2024–25	Jahnavi Kolakaluri, MS-HCI , Independent Study on "Noondawind: Data Storytelling" & Graduate Research Assistantship on "Kumu Connect" and "Noondawind"
2024–25	Michael Parkin, MS-HCI , Graduate Research Assistantship on "Kaona"